

Abdul Basit Khan

Software Engineering Student | Full-Stack & AI/ML Developer

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Portfolio: [abdulbasitkhan1947.github.io/abdulbasit-portfolio](https://github.com/abdulbasitkhan1947)

PROFESSIONAL SUMMARY

Results-driven final-year Software Engineering student (PAF-IAST, graduating 2025) with three industry internships spanning full-stack web engineering, AI/ML model development, and 3D game development. Architect of HealthTrack AI — a production-grade, end-to-end healthcare platform integrating clinical NLP (BioBERT/ClinicalBERT), TOPSIS doctor-matching, real-time telemedicine, and cross-platform mobile (Flutter) — built with Next.js 14, Node.js, PostgreSQL, MongoDB, and Hugging Face Transformers on a 189 K-row medical dataset. Adept at translating complex business problems into deployable AI and web solutions. Actively seeking an internship or junior developer role where I can contribute from day one.

TECHNICAL SKILLS

Languages	JavaScript, Python, Java, C++, SQL
Frontend	React, Next.js 14, TypeScript, HTML5, CSS3, Tailwind CSS, Bootstrap, ShadCN UI
Backend	Node.js, Express.js, Flask
Databases	PostgreSQL, MongoDB, MySQL, Redis
AI / ML	Hugging Face Transformers, scikit-learn, TensorFlow.js, spaCy, PyTorch (basic), Dialogflow
Mobile	Flutter 3.0 (cross-platform, iOS & Android)
DevOps & Tools	Git, GitHub, Docker (basic), Vercel, Railway, Postman, VS Code
Specialisations	NLP / clinical NLP, real-time systems (Socket.io), GIS integration, Merkle-tree audit trails

WORK EXPERIENCE

AI/ML Engineer Intern — Arch Technologies, Haripur 2025

- Designed and trained classification and regression ML models in Python (scikit-learn / TensorFlow), translating raw business requirements into deployable AI solutions that improved decision accuracy.
- Engineered end-to-end data preprocessing pipelines — cleaning, normalising, and feature-engineering datasets — directly increasing model training efficiency and reducing data-quality errors by standardising ingestion across 3+ data sources.
- Bridged communication between technical teams and non-technical stakeholders by converting real-world business problems into actionable ML problem statements, shortening project scoping cycles.

Web Engineer Intern — Codeque Software Solutions, Haripur 2023

- Developed and shipped responsive web features using HTML5, CSS3, and JavaScript, ensuring pixel-accurate rendering across 3+ major browsers (Chrome, Firefox, Safari) with zero regressions.
- Executed systematic debugging and QA testing across the frontend codebase, identifying and resolving root-cause defects and reducing bug recurrence.
- Managed Git version-control workflows — branching, pull requests, code reviews — enabling clean collaborative development within the team.

Digital Game Developer Intern — M-Labs, Lahore 2024

- Engineered immersive 3D game environments in Unity, implementing physics, lighting, and spatial design systems across multiple interactive scenes.
- Architected event-driven game mechanics with clean separation of game state, events, and UI, enabling responsive and maintainable gameplay systems.

KEY PROJECTS

HealthTrack AI — *Final-Year Project | Next.js 14 · Node.js · Flutter · BioBERT · PostgreSQL · MongoDB*

- Architected a full-stack, AI-powered healthcare platform supporting multi-modal symptom input (text, voice, image) via clinical NLP models (BioBERT/ClinicalBERT) trained on a 189,034-row, 322-feature medical dataset — delivering intelligent, context-aware doctor recommendations.
- Implemented a TOPSIS multi-criteria decision algorithm with GIS integration to rank 40+ doctors by qualification, specialisation, and real-time availability, achieving transparent and explainable recommendations.
- Delivered real-time telemedicine (ZegoCloud), a blockchain-inspired Merkle-tree health vault, offline-first Android support, and bilingual (Urdu/English) UX; deployed across Vercel (frontend), Railway (backend), and Supabase (DB).

AI Symptom Extractor (NLP Microservice) — *Python · spaCy · Hugging Face · Flask*

- Built a production-ready NLP microservice that extracts structured medical symptoms from free-text patient input, integrating spaCy and Hugging Face Transformers for high-accuracy entity recognition.
- Exposed as a plug-and-play Flask REST API, enabling seamless integration with chatbots and telemedicine platforms; directly powers the symptom-analysis pipeline in HealthTrack AI.

Dispensary Management System — *Node.js · Express · MongoDB · HTML/CSS*

- Designed and delivered a full-stack clinic-operations platform managing patient records, medicine inventory, and appointment scheduling, eliminating manual record-keeping.
- Built a RESTful backend with Node.js, Express, and MongoDB, applying normalised data modelling to handle concurrent patient and inventory transactions reliably.

School LMS & Library Management System — *Web-based, full-stack*

- Developed two institutional web platforms supporting course management, student tracking, book cataloguing, and borrowing workflows, serving multiple role types (admin, student, librarian) within a single system.

EDUCATION

BS Software Engineering

Graduating: July 2026

PAF-IAST (Pak-Austria Fachhochschule), Haripur | CGPA: 2.62 / 4.0 | Final Semester

Relevant coursework: Software Development Life Cycle (Agile/Waterfall), Data Structures & Algorithms, Databases (SQL), OOP (Java, C++), Machine Learning, Web Engineering

ACHIEVEMENTS & ACTIVITIES

- Led cross-functional university project teams across 5+ academic projects delivered on schedule.
- Completed 3 industry internships across AI/ML, full-stack web, and game development during undergraduate studies.
- Published live portfolio at abdulbasitkhan1947.github.io/abdulbasit-portfolio showcasing all major projects.

LANGUAGES

English — Professional Proficiency | Urdu — Native